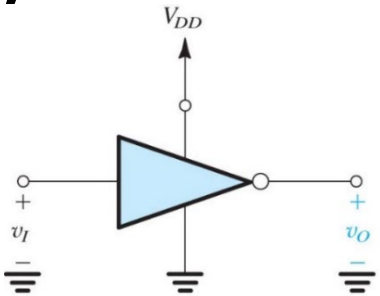
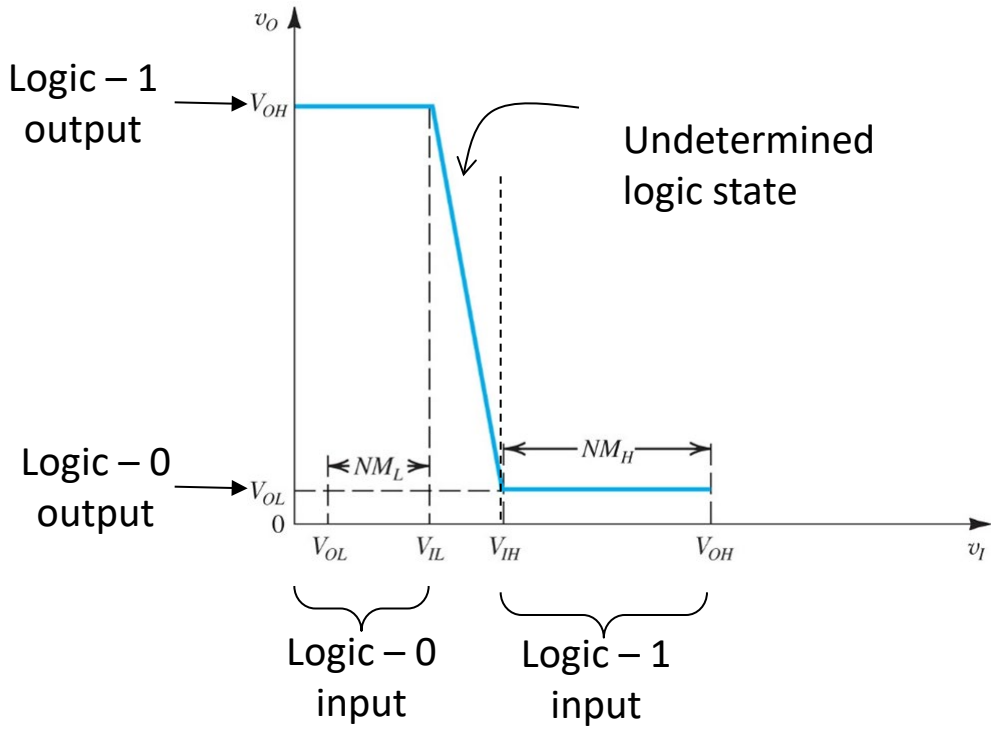
Unit #2
Chapter #16.2: Digital Logic Inverters
Lecture 7

from **Microelectronic Circuits** Text
by Sedra and Smith
Oxford Publishing

16.2.1 **The Voltage Transfer Characteristic (VTC)**

- The inverter "inverts" the logic value of input (v_I) to create the opposite logic value at output (v_O)



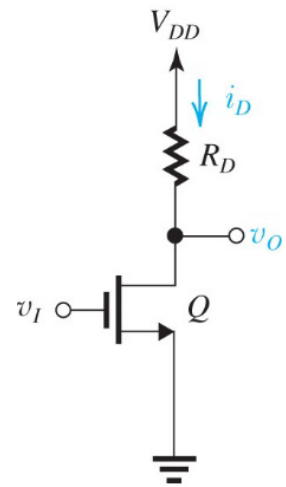


v_I	v_O
Logic – 0	Logic – 1
Logic – 1	Logic – 0

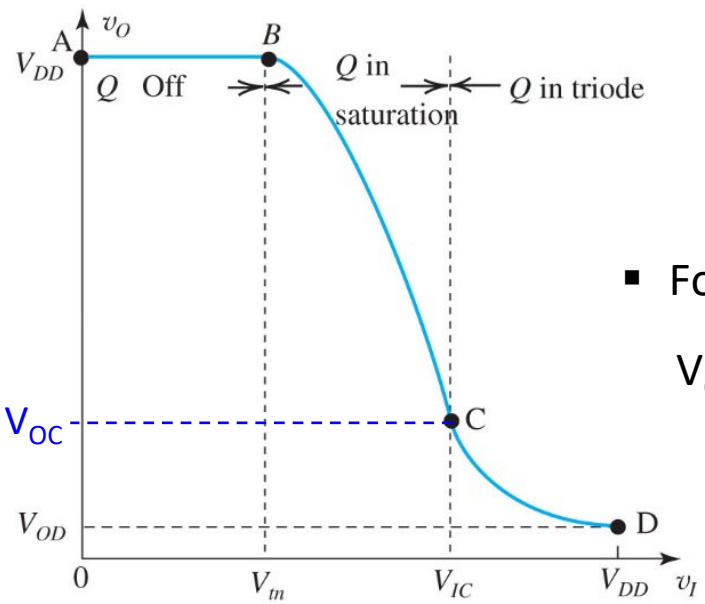
v_I	v_O
F	T
T	F

16.2.1 **Simple NMOS Inverter**

- A simple inverter is just an NMOS transistor in series with a drain resistor



- VTC for simple NMOS inverter



- For NMOS inverter

  $V_{OC} = V_{IC} - V_{tn}$ (edge of sat/triode)

16.2.1 **Range of Logic Levels (Input)**

- Logic-0 if $v_I \le V_{IL}$ where V_{IL} is the highest input that produces a logic-1 output
- Logic-1 if $V_{IH} \le v_I$ where V_{IH} is the lowest input that produces a logic-0 output

For NMOS inverter

$$V_{IL} \approx V_{tn}$$

$$V_{IH} = V_{OC} + V_{tn}$$

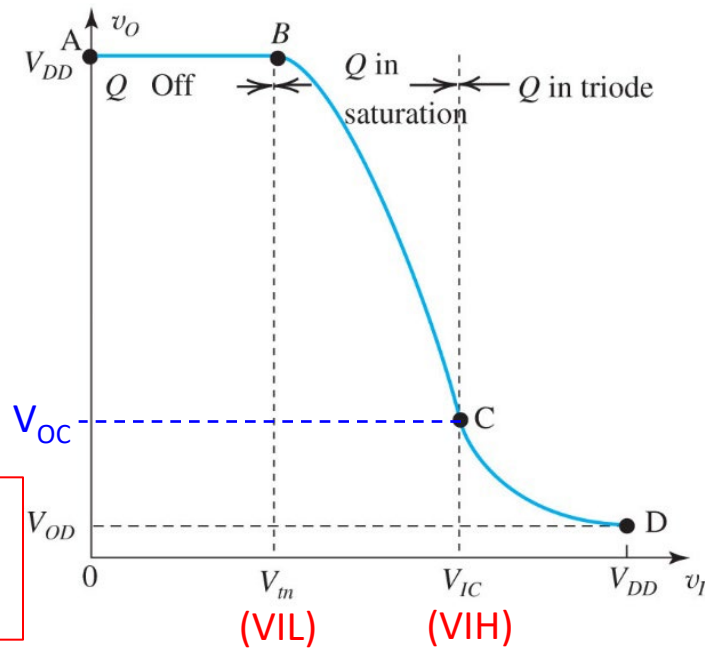
Edge of Sat. & Triode

where

$$V_{OC} = V_{OV}$$

and

$$\frac{V_{DD} - V_{OV}}{R_D} = \frac{k_n}{2} V_{OV}^{\ 2}$$



16.2.1 **Range of Logic Levels (Output)**

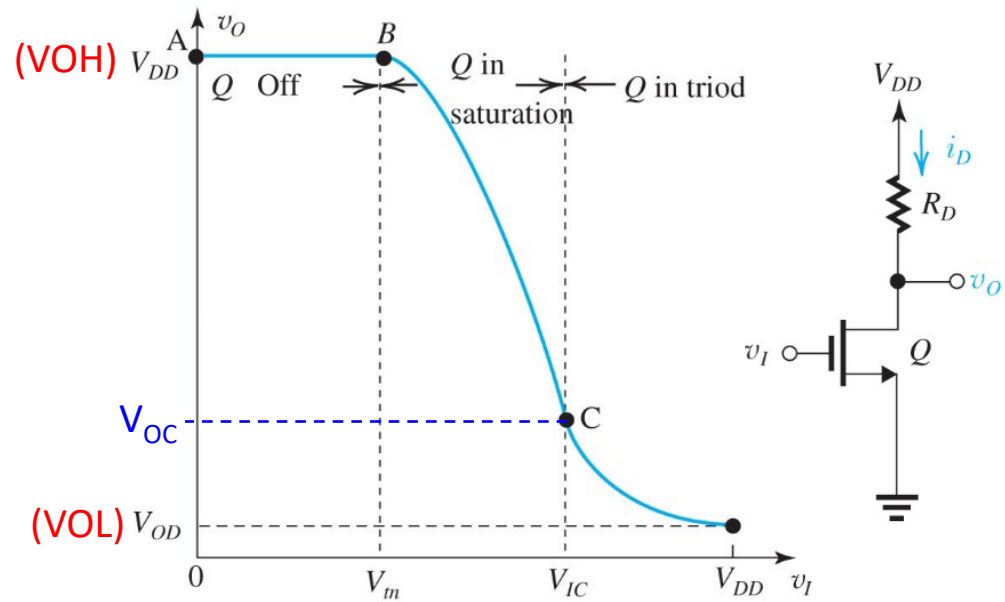
- Logic-0 if $V_{OL} \le v_O \le V_{OC}$ where V_{OL} is the lowest output, at $v_I = V_{DD}$
- Logic-1 if $V_{OB} \le v_O \le V_{OH}$ where V_{OH} is the highest output, at $v_I = 0V$ … i.e., $V_{OH} = V_{DD}$

For NMOS inverter

$$V_{OH} = V_{OB} = V_{DD}$$

V_{OL} is the solution of

$$k_n\left((V_{DD} - V_{tn})V_{OL} - \frac{V_{OL}^2}{2}\right) = \frac{V_{DD} - V_{OL}}{R_D}$$



Important Note: There is current (power dissipated) when input equals V_{DD}.

16.2.2 **Noise Margins**

- Figure of merit for inverter
- Determines the amount of noise the inverter can tolerate
- There are high and low noise margins

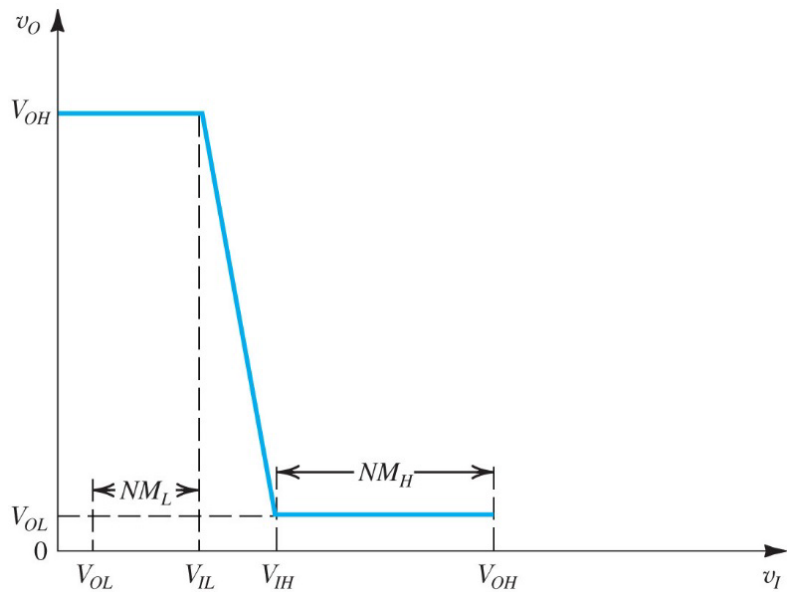
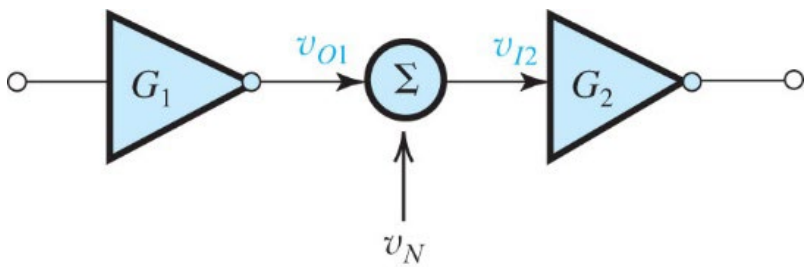


Table 14.1 Important Parameters of the VTC of the Logic Inverter (Refer to Fig. 14.13)

V_{OL}:	Output low level
V_{OH}:	Output high level
V_{IL}:	Maximum value of input interpreted by the inverter as a logic 0
V_{IH}:	Minimum value of input interpreted by the inverter as a logic 1
NM_L:	Noise margin for low input $= V_{IL} - V_{OL}$
NM_H:	Noise margin for high input $= V_{OH} - V_{IH}$

16.2.2 **Noise Margin (Explained)**

- Consider a chain of two inverters and a noise source inserter between output of 1st and input of 2nd



- If the output of the 1st is V_{OL} then we can tolerate at most

$$V_{IL} = V_{OL} + v_{N(\max)}$$

so

$$v_{N(\max)} = V_{IL} - V_{OL} = N_{ML}$$

- If the output of the 1st is V_{OH} then we can tolerate at most

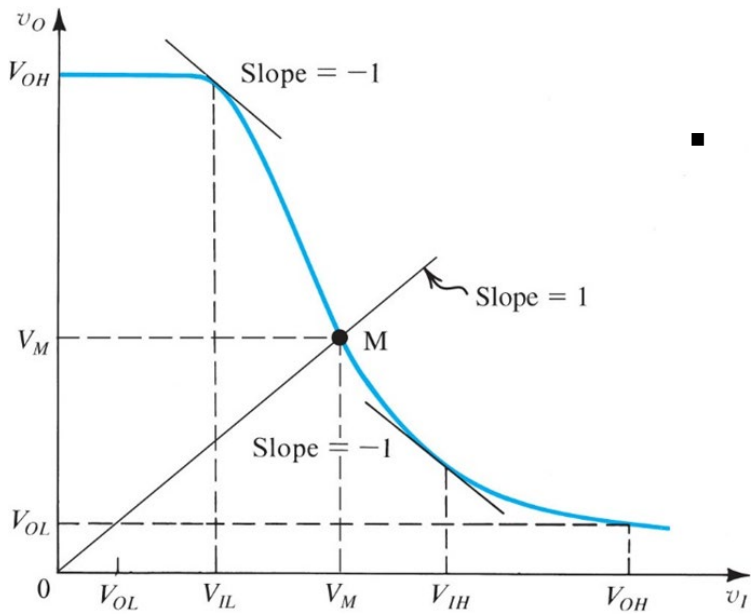
$$V_{IH} = V_{OH} - v_{N(\max)}$$

so

$$v_{N(\max)} = V_{OH} - V_{IH} = N_{MH}$$

16.2.2 **Typical Critical Point Definition**

- While V_{OL} and V_{OH} are well defined, we need a more specific definition for V_{IL} and V_{IH}
- Typically V_{IL} and V_{IH} are defined as the VTC points at which the slope is -1 V/V



- Note the midpoint of the VTC is defined as the point where

$$v_I = v_O = V_M$$

MC74LCX04

Low-Voltage CMOS Hex Inverter

With 5 V−Tolerant Inputs

The MC74LCX04 is a high performance hex inverter operating from a 2.0 to 5.5 V supply. High impedance TTL compatible inputs significantly reduce current loading to input drivers while TTL compatible outputs offer improved switching noise performance. A V_I specification of 5.5 V allows MC74LCX04 inputs to be safely driven from 5 V devices if V_{CC} is less than 5.0 V.

Current drive capability is 24 mA at the outputs.

Features

- Designed for 2.0 V to 5.5 V V_{CC} Operation
- 5 V Tolerant Inputs − Interface Capability With 5 V TTL Logic
- LVTTL Compatible
- LVCMOS Compatible
- 24 mA Balanced Output Sink and Source Capability
- Near Zero Static Supply Current (10 μA) Substantially Reduces System Power Requirements
- Latchup Performance Exceeds 500 mA
- ESD Performance: Human Body Model >2000 V; Machine Model >200 V
- NLV Prefix for Automotive and Other Applications Requiring Unique Site and Control Change Requirements; AEC−Q100 Qualified and PPAP Capable
- These Devices are Pb−Free, Halogen Free/BFR Free and are RoHS Compliant

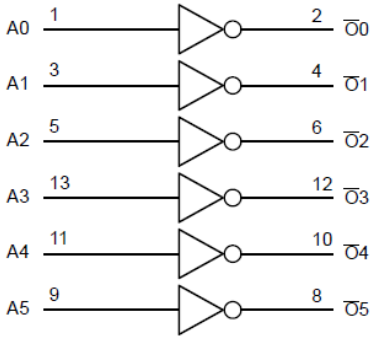


Figure 1. Logic Diagram

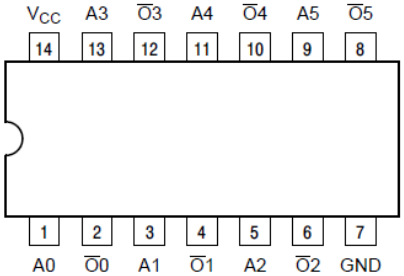
ON Semiconductor®

http://onsemi.com

SOIC−14 NB
D SUFFIX
CASE 751A

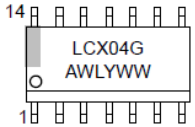
TSSOP−14
DT SUFFIX
CASE 948G

PIN ASSIGNMENT

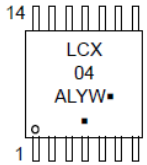


14−Lead (Top View)

MARKING DIAGRAMS



SOIC−14 NB



TSSOP−14

A = Assembly Location
WL, L = Wafer Lot
YY, Y = Year

MC74LCX04

DC ELECTRICAL CHARACTERISTICS

Symbol	Characteristic	Condition	T_A = −55°C to +125°C Min	Max	Unit
V_{IH}	HIGH Level Input Voltage (Note 2)	2.3 V ≤ V_{CC} ≤ 2.7 V	1.7		V
		2.7 V ≤ V_{CC} ≤ 3.6 V	2.0		
V_{IL}	LOW Level Input Voltage (Note 2)	2.3 V ≤ V_{CC} ≤ 2.7 V		0.7	V
		2.7 V ≤ V_{CC} ≤ 3.6 V		0.8	
V_{OH}	HIGH Level Output Voltage	2.3 V ≤ V_{CC} ≤ 3.6 V; I_{OH} = −100 μA	V_{CC} − 0.2		V
		V_{CC} = 2.3 V; I_{OH} = −8 mA	1.8		
		V_{CC} = 2.7 V; I_{OH} = −12 mA	2.2		
		V_{CC} = 3.0 V; I_{OH} = −18 mA	2.4		
		V_{CC} = 3.0 V; I_{OH} = −24 mA	2.2		
V_{OL}	LOW Level Output Voltage	2.3 V ≤ V_{CC} ≤ 3.6 V; I_{OL} = 100 μA		0.2	V
		V_{CC} = 2.3 V; I_{OL} = 8 mA		0.6	
		V_{CC} = 2.7 V; I_{OL} = 12 mA		0.4	
		V_{CC} = 3.0 V; I_{OL} = 16 mA		0.4	
		V_{CC} = 3.0 V; I_{OL} = 24 mA		0.55	
I_{OFF}	Power Off Leakage Current	V_{CC} = 0, V_{IN} = 5.5 V or V_{OUT} = 5.5 V		10	μA
I_{IN}	Input Leakage Current	V_{CC} = 3.6 V, V_{IN} = 5.5 V or GND		±5	μA
I_{CC}	Quiescent Supply Current	V_{CC} = 3.6 V, V_{IN} = 5.5 V or GND		10	μA
ΔI_{CC}	Increase in I_{CC} per Input	2.3 ≤ V_{CC} ≤ 3.6 V; V_{IH} = V_{CC} − 0.6 V		500	μA

2. These values of V_I are used to test DC electrical characteristics only.